Onboarding

System Description

Abstract

This document describes the Onboarding support system of the Eclipse Arrowhead 5.2.2 framework, as implemented in `ah5-Onboarding-java-spring` of the primary Java source tree (`Eclipse-Arrowhead-java`) and deployed with the standalone Compose definition of that folder.

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1 Overview

This document describes the Onboarding system. Onboarding implements the Arrowhead 5.2.2 onboarding procedure: a device holding only a manufacturer-issued bootstrap certificate, approved by an operator, obtains its onboarding, device and system certificates from the local-cloud CertificateAuthority, its ServiceRegistry device entry and the Authentication identities of its systems. Off-boarding withdraws all of it.

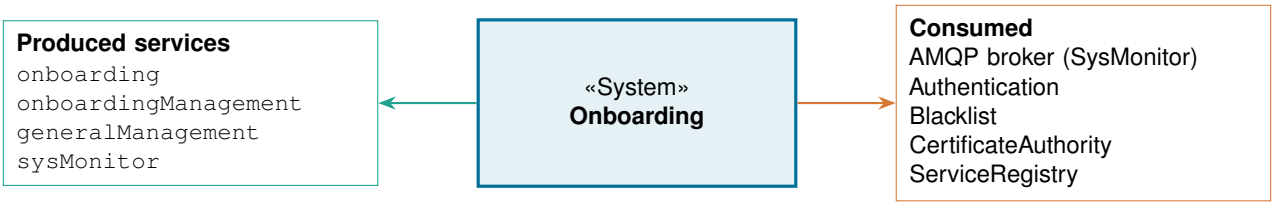


Figure 1: SysML-style block view of Onboarding with its produced and consumed services.

The rest of this document is organized as follows. In Section 1.1, we reference major prior art capabilities of the system. In Section 1.2, we describe the intended usage of the system. In Section 1.3, we describe fundamental properties provided by the system. In Section 1.4, we describe delimitations of capabilities of the system. In Section 2, we describe the abstract service functions consumed or produced by the system. In Section 3, we describe the security capabilities of the system.

1.1 Significant Prior Art

After Bicaku et al., Interacting with the Arrowhead Local Cloud: On-boarding Procedure (2018); migration of the v4.6.2 onboarding controller.

1.2 How This System Is Meant to Be Used

Operator: `approve-device` with the bootstrap fingerprint and the approved systems. Device: `onboard-with-csr`, `onboard-device`, then `onboard-system` per system, each step with the certificate of the previous one. Systems then log in, register and orchestrate. `offboard-device` (device or operator) removes everything, revokes the certificates and blacklists the systems.

1.3 System functionalities and properties

1.3.1 Functional properties of the system

- Certificate-only enrollment with approved bootstrap certificates (SHA-256 fingerprint).
- Onboarding (`on`, at most 24 h), device (`de`) and system (`sy`) certificates signed by the CertificateAuthority; CSR or generated key.
- ServiceRegistry device entry and Authentication identities.
- Off-boarding: registry entries, identities, revocation, blacklisting.

1.3.2 Configuration of system properties

The system is configured with Spring properties (`application.properties`, overridden by environment variables in the container deployment). The properties that shape its behaviour are:

Property	Default	Meaning
`server.port`	8435	HTTPS port.
`authentication.policy`	outsourced	Identity tokens for generalManagement.
`service.interface.policy`	CERT_AUTH	Policy of the registered interfaces.
`management.policy`	sysop-only	Access to generalManagement.
`ONBOARDING_TLS_*`, `ONBOARDING_TRUSTSTORE*`	(required)	System certificate and the trusted cloud and manufacturer CAs.
`ONBOARDING_HOST`	localhost	Advertised host.
`ONBOARDING_DB_USER`	onboarding	Database user (password from a secret file).
`ONBOARDING_RESERVED_NAMES`	core, support and operator system names	Names that are never approved for a device or an onboarded system.
`sysmonitor.*`	true	Native `sysMonitor` publisher.
`arrowhead.ssl.crl-file`	(unset)	CRLs of the cloud CA and the manufacturer CAs.

Secrets (database, keystore and identity passwords, secret keys) have no usable defaults; the standalone deployment reads them from per-deployment Compose file secrets under `/run/secrets`.

The standalone deployment of this folder uses `config/application.properties` (copied into the image) with `server.port`=8435, `authentication.policy=outsourced`, `service.interface.policy=CERT_AUTH`, `management.policy=sysop-only`, `service.authorization.enabled=true`, `mqtt.api.enabled=false`, `arrowhead.ssl.crl-file=/crl/ca.crl`; secrets are read from Compose file secrets under `/run/secrets` (`spring.config.import=configtree:/run/secrets/`). `enable.`

`blacklist.filter` is not enabled there, so incoming requests are not checked against the Blacklist; blacklisted providers are still removed by DynamicServiceOrchestration.

1.3.3 Data stored by the system

Device and system enrollments and log entries (database `ah_onboarding`). No CA key.

1.3.4 Non functional properties

Each step calls the CertificateAuthority, the ServiceRegistry and Authentication synchronously.

1.3.5 Stateful or stateless

Stateful: enrollments are persistent.

1.4 Important Delimitations

Onboarding does not sign certificates itself and does not approve devices; an operator does. An approval names 1 to 32 systems per device, none of them a reserved name (`ONBOARDING_RESERVED_NAMES`: the core and support systems, Sysop and the test clients). Only the onboarding and device certificate issued last are accepted, and only while the CA reports them GOOD.

2 Services

This section describes the produced and consumed services. Each produced service is specified by an SD document and one IDD document per interface.

2.1 Produced services

Service	Version	Interfaces	SD	IDD
`onboarding`	1.0.0	HTTP(S)	[1]	[2]
`onboardingManagement`	1.0.0	HTTP(S)	[3]	[4]
`generalManagement`	1.0.0	HTTP(S)	[5]	[6]
`sysMonitor`	1.0.0	AMQP(S)	[7]	[8]

The HTTP interfaces are always offered and the AMQP `sysMonitor` interface with `sysmonitor.enabled=true`. The unregistered endpoints of the implementation are: `GET /onboarding/echo`.

2.2 Consumed services

Provider	Service	Operations	Use	Condition
ServiceRegistry	systemDiscovery, serviceDiscovery	register, revoke, lookup	Registration of the system and its services; lookup of the bootstrap services.	always
ServiceRegistry	serviceRegistry Management	device-create, device-update, device-remove, system-remove	Device and system entries.	management whitelist
Authentication	identity	identity-login, identity-verify	Own identity token; verification of requester tokens.	`authentication.policy=outsourced`
Authentication	identity Management	identity-mgmt-create, identity-mgmt-remove	System identities.	grant
Certificate Authority	certificateSigning; certificate Validation; certificateAuthority Management	sign; check-certificate; revoke-certificate	Certificates.	grant
Blacklist	blacklist Management	create, remove	Blacklisting of offboarded systems.	grant
AMQP broker (SysMonitor)	sysMonitor (publisher)	echo, power, status	Own observations.	`sysmonitor.enabled=true`

Except for the bootstrap services, which are looked up in the ServiceRegistry, the system reaches other systems' services through DynamicServiceOrchestration (`http.collector.mode=ORCHESTRATION`, results cached for 300 s); every such call needs a ConsumerAuthorization grant.

3 Security

The system can run in an insecure mode (`server.ssl.enabled=false`, plain HTTP, identity declared by the requester) for development, and in the Arrowhead secure mode with HTTPS and X.509 certificates following the Arrowhead certificate profile (`<System>.<Cloud>.<Organization>.arrowhead.eu`). Certificates not issued under the cloud CA are refused by the TLS handshake. The standalone deployment of this folder (`config/application.properties` and `compose.yaml`) runs the secure mode.

3.1 Security model

- **Protocols:** HTTPS (TLS 1.3 and 1.2) with mandatory client certificates; MQTT over TLS; AMQP over TLS for `sysMonitor`.
- **Data protection:** transport encryption by TLS; the database connection uses TLS with server verification in the standalone deployment. Secrets are kept out of the container environment (Compose file secrets).
- **System authentication:** by the authentication policy: `certificate`, `outsourced` (identity token verified with Authentication) or `declared` (insecure); the standalone deployment uses `outsourced`. An identity token is accepted only together with the client certificate of the system it was issued to (token-to-certificate binding), and every client certificate is checked against the cloud CA's CRL (fail closed).
- **Authorization of produced services:** management services are protected by the management filter (`management.policy`: `SYSOP_ONLY`, `WHITELIST` or `AUTHORIZATION`); with `service.authorization.enabled=true` ConsumerAuthorization decides each call; an optional blacklist filter refuses blacklisted requesters. Sysop identities bypass the management filter.
- Every caller presents a trusted client certificate; certificates and peer IP are taken from the TLS connection (no forwarded identity headers). `onboardingManagement` needs the Sysop token bound to the operator certificate.

For the Arrowhead certificate profile see github.com/eclipse-arrowhead/documentation.

4 References

[1] J. Delsing, "onboarding – Service Description (Onboarding)," Eclipse Arrowhead, SD 5.2.2, 2026, onboarding-v5.2.2-onboarding-SD.pdf.

[2] ——, "onboarding HTTP/TLS/JSON – Interface Design Description (Onboarding)," Eclipse Arrowhead, IDD 5.2.2, 2026, onboarding-v5.2.2-onboarding-HTTP-IDD.pdf.

[3] ——, "onboardingManagement – Service Description (Onboarding)," Eclipse Arrowhead, SD 5.2.2, 2026, onboarding-v5.2.2-onboardingManagement-SD.pdf.

[4] ——, "onboardingManagement HTTP/TLS/JSON – Interface Design Description (Onboarding)," Eclipse Arrowhead, IDD 5.2.2, 2026, onboarding-v5.2.2-onboardingManagement-HTTP-IDD.pdf.

[5] ——, "generalManagement – Service Description (Onboarding)," Eclipse Arrowhead, SD 5.2.2, 2026, onboarding-v5.2.2-generalManagement-SD.pdf.

[6] ——, "generalManagement HTTP/TLS/JSON – Interface Design Description (Onboarding)," Eclipse Arrowhead, IDD 5.2.2, 2026, onboarding-v5.2.2-generalManagement-HTTP-IDD.pdf.

[7] ——, "sysMonitor – Service Description (Onboarding)," Eclipse Arrowhead, SD 5.2.2, 2026, onboarding-v5.2.2-sysMonitor-SD.pdf.

[8] ——, "sysMonitor AMQP/TLS/JSON – Interface Design Description (Onboarding)," Eclipse Arrowhead, IDD 5.2.2, 2026, onboarding-v5.2.2-sysMonitor-AMQP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated; database user and reserved-name settings added.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	